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Solution

$L_{air} = 4.8 \times 10^{-3} H$

$L_{iron} = 1.8H$

$L_{air} = \mu_{\circ} n^2 Al = 4.8 \times 10^{-3} H$

$L_{iron} = \mu n^2 Al = \mu_{\circ} \mu_r n^2 Al = 1.8H$

$$\therefore \mu_r = \frac{L_{iron}}{L_{air}} = \frac{1.8}{4.8 \times 10^{-3}} = 375$$

4.3.3 Mutual induction

When an electric current passing through a coil changes with time, an emf is induced in the neighbouring coil. This phenomenon is known as mutual induction and the emf is called mutually induced emf.

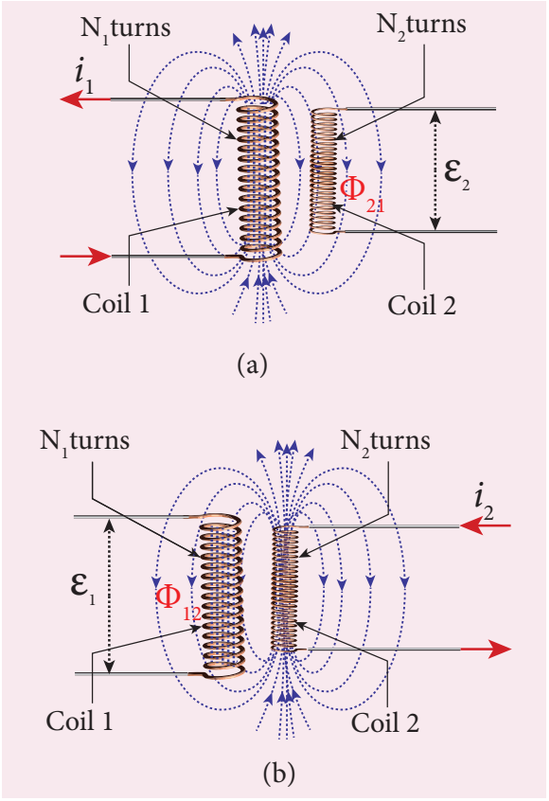


Figure 4.22 Mutual induction

Consider two coils which are placed close to each other. If an electric current i_1 is sent through coil 1, the magnetic field produced by it is also linked with coil 2 as shown in Figure 4.22(a).

Let Φ_{21} be the magnetic flux linked with each turn of the coil 2 of N_2 turns due to coil 1, then the total flux linked with coil 2 $(N_2 \Phi_{21})$ is proportional to the current i_1 in the coil 1.

$$N_2 \Phi_{21} \propto i_1$$

$$N_2 \Phi_{21} = M_{21}\, i_1 \qquad (4.19)$$

$$\text{(or)}\ M_{21} = \frac{N_2 \Phi_{21}}{i_1}$$

The constant of proportionality M_{21} is the mutual inductance of the coil 2 with respect to coil 1. It is also called as coefficient of mutual induction. If $i_1 = 1A$, then $M_{21} = N_2 \Phi_{21}$. Therefore, **the mutual inductance M_{21} is defined as the flux linkage of the coil 2 when 1A current flows through coil 1.**

When the current i_1 changes with time, an emf ε_2 is induced in coil 2. From Faraday's law of electromagnetic induction, this mutually induced emf ε_2 is given by

$$\varepsilon_2 = -\frac{d\left(N_2 \Phi_{21}\right)}{dt} = -\frac{d(M_{21} i_1)}{dt}$$

$$\varepsilon_2 = -M_{21} \frac{di_1}{dt}$$

$$\text{(or)}\ M_{21} = \frac{-\varepsilon_2}{di_1 / dt}$$

The negative sign in the above equation shows that the mutually induced emf always opposes the change in current i_1 with respect to time. If $di_1/dt = 1A\ s^{-1}$, then $M_{21} = -\varepsilon_2$.

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